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Mohamed Ahmed Ahmed Shams

Email: mohamed.ah.shams@gmail.com | Portfolio

Professional Summary

Systems Engineer with hands-on experience delivering large-scale SCADA and Industrial IoT solutions for government and industrial clients. Strong background in Ignition SCADA, distributed system architectures, industrial communication protocols, and database-driven applications. Experienced across the full project lifecycle including system design, deployment, integration, commissioning, and user enablement, with growing expertise in data analytics and AI-enabled industrial solutions.

Projects

1. **NCEC Air Quality Monitoring System / AQMS, 04/2026 - Present**
   Supported the implementation of the AQMS project using Stream IIOT Platform and MSSQL, focusing on station data configuration, database support, platform validation, and reliable visualization of air quality monitoring data.
2. **DMAs / DMZs (Network Improvement Works) for Northwest Cluster, 03/2026 – 07/2026**
   Delivering RTU, and testing activities for NWC water network improvement project across Madinah, Yanbu, and Tabuk.
3. **Döhler Factory Cloudy Line, 10/2024 - 02/2026**
   Developed the complete Ignition SCADA system for Döhler Egypt's new production line producing 65 Brix cloudy juice concentrate from orange peel waste.
4. **MODON Industrial Smart Cities Platform & Mobile App Testing, 11/2025 - 02/2026**
   Conducted end-to-end manual testing for MODON web and mobile journeys with Azure DevOps, REST APIs, regression, compatibility, and acceptance criteria validation.
5. **Ground Water Monitoring Platform, 09/2024 - 12/2025**
   Contributed to a scalable Ignition architecture monitoring over 7,000 underground wells with Oracle DB, InfluxDB, REST APIs, dashboards, and role-based access.
6. **Underground Wells Drilling Equipment Tracking Platform, 09/2024 - 12/2025**
   Developed an Ignition-based governance and monitoring platform for 1,300 drilling equipment assets across underground wells.
7. **MODON Industrial Smart Cities Platform (POC), 06/2024 - 11/2024**
   Assisted in developing a proof-of-concept Ignition platform to monitor over 10,000 smart sensors across industrial cities.

Technologies/Technical Skills

- Ignition SCADA
- AVEVA Application Server
- WinCC
- SCADA & HMI Engineering
- Industrial IoT & Smart Infrastructure
- MS SQL Server / MySQL / PostgreSQL
- InfluxDB time-series historian
- OPC-UA / MQTT / REST APIs
- Python
- JavaScript / HTML / CSS
- Docker / Git / n8n
- Dataiku workflows
- AWS AI services foundations

Certificates/Accreditation

- AVEVA Application Server - Apr 2026
  AVEVA Learning Academy
- AWS Certified AI Practitioner - Feb 2026
  Amazon Web Services
- Dataiku ML Practitioner Certified - Dec 2025
  Dataiku
- Dataiku Core & Advanced Designer Certified - Sep 2025
  Dataiku
- Ignition Core Certified - Nov 2024
  Inductive Automation

Training Courses

- Data Science & AI Course - Jun-Nov 2025
  Machinfy
- GEN-110: Cloudera Essentials - Nov 2025
  Cloudera
- Containerization & Virtualization with Docker & Kubernetes - May 2025
  DataCamp
- Complete RAG System: From LLM Retrieval to AWS Deployment - Apr 2025
  Udemy
- WinCC Unified Essentials - Aug 2024
  Siemens Unified Academy
- IIOT Courses: MQTT, OPC, Node-RED, PLC Integrations - May 2024
  Codeandcompile
- Advanced & Basic Automation Courses - Feb 2023
  HA Consulting Group

Professional Experience

pulse by solutions **Sep 2024 - Present**
Systems Engineer

Project Name: NCEC Air Quality Monitoring System
From 04/2026 till now
Client: National Center for Environmental Compliance (NCEC), Saudi Arabia
Job Title: Systems Engineer (pulse by solutions)
Applications: Stream IIOT Platform, MSSQL
Overview: Supported the implementation of the NCEC AQMS project using Stream IIOT Platform and MSSQL, focusing on station data, database support, platform development of air quality monitoring data.
Responsibilities:
- Configured and supported AQMS data on the Stream IIOT Platform.
- Worked with MSSQL queries and structured station/analyzer data.
- Validated platform data, displayed values, and system behavior.
- Supported troubleshooting, testing, and project documentation.

Project Name: DMAs / DMZs (Network Improvement Works) for Northwest Cluster
From 03/2026 till 07/2026
Client: National Water Company (NWC), Saudi Arabia
Job Title: Systems Engineer (pulse by solutions)
Applications: SOFREL RTU/Datalogger
Overview: Delivering RTU, and testing activities for NWC's DMAs / DMZs water network improvement project across Madinah, Yanbu, and Tabuk.
Responsibilities:
- Delivering SOFREL RTU/datalogger works for field sites.
- Performing signal, alarm, and communication validation.
- Preparing FAT, T&C, and other technical project documentation.
- Executing testing and commissioning activities.

Project Name: Döhler Factory Cloudy Line - First Cloudy Line in the Middle East
From 10/2024 till 02/2026
Client: Döhler Egypt
Job Title: Systems Engineer (pulse by solutions)
Applications: Ignition, PostgreSQL
Overview: Developed the complete SCADA system for Döhler Egypt's new production line producing 65 Brix cloudy juice concentrate from orange peel waste, the first of its kind in the Middle East.
Responsibilities:
- Developed a complete SCADA system using Ignition for a new cloudy juice concentrate line.
- Designed dynamic, responsive HMIs using the Perspective module, integrated with MSSQL for historians.
- Connected and monitored PLCs from 7+ package suppliers.
- Implemented alarm monitoring and live visualization of sensors, valves, VSDs, and process parameters.
- Supported commissioning, operator training, and alignment with the factory's digital systems.

Project Name: MODON Industrial Smart Cities Platform & Mobile App Testing
From 11/2025 till 02/2026
Client: MODON (Saudi Authority for Industrial Cities and Technology Zones)
Job Title: Systems Engineer (pulse by solutions)
Applications: Azure DevOps, Web Platform, Mobile App, REST APIs
Responsibilities:
- Conducted end-to-end manual testing for the MODON Web Portal and Mobile App across all user journeys.
- Verified core features including authentication, complaints, and dashboards in Azure DevOps, ensuring 100% alignment with acceptance criteria.
- Performed cross-browser, device compatibility, and regression testing to maintain stable, high-quality production releases.

Project Name: Ground Water Monitoring Platform
From 09/2024 till 12/2025
Client: Saudi Arabia's Ministry of Environment, Water and Agriculture (MEWA)
Job Title: Systems Engineer (pulse by solutions)
Applications: Ignition, Oracle DB, InfluxDB
Responsibilities:
- Contributed to the deployment of a scalable Ignition architecture monitoring over 7,000 underground wells for real-time water consumption analysis.
- Participated in scale-out architecture design involving multiple gateways with secure network and redundancy setup.
- Assisted in designing and implementing the Oracle database schema to support system applications.
- Configured InfluxDB as a time-series historian for Ignition data.
- Supported integration with third-party systems via REST APIs.
- Helped implement Active Directory integration and role-based user management.
- Contributed to asset management features and creation of analytical dashboards.

Project Name: Underground Wells Drilling Equipment Tracking Platform
From 09/2024 till 12/2025
Client: Saudi Arabia's Ministry of Environment, Water and Agriculture (MEWA)
Job Title: Systems Engineer (pulse by solutions)
Applications: Ignition, Oracle DB, InfluxDB
Responsibilities:
- Developed an Ignition-based platform for governance and monitoring of 1,300 drilling equipment assets across underground wells.
- Assisted with designing the distributed system architecture across multiple gateways with secure communication and failover mechanisms.
- Participated in Oracle DB schema design for system-wide data management.

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- Configured InfluxDB as a time-series historian for Ignition data.
- Helped with building asset registration workflows and interactive dashboards for data presentation.
- Supported integration with third-party systems via REST APIs.
- Helped implement Active Directory integration and role-based user management.

Project Name: MODON Industrial Smart Cities Platform (POC)
From 06/2024 till 11/2024
Client: MODON (Saudi Authority for Industrial Cities and Technology Zones)
Job Title: Systems Engineer (pulse by solutions)
Applications: Ignition, MSSQL
Responsibilities:

- Assisted in developing a proof-of-concept Ignition platform to monitor over 10,000 smart sensors deployed across industrial cities.
- Supported system architecture design using an All-in-One setup with secure network configurations.
- Helped design the MSSQL database scheme for application integration and data management.
- Contributed to third-party API integration and data exchange.
- Supported asset registration processes and creation of data dashboards.

Education

B.Sc., Electrical Engineering - Menoufia University - July 2024

Languages

Arabic: Mother Tongue
English: Very Good

Personal Data

Birth: 1st October 2001
Nationality: Egyptian
Gender: Male

References

Furnished upon request.